

THE SOLAR SYSTEM

THE SOLAR SYSTEM FORMED FROM A SLOWLY ROTATING NEBULA OF GAS AND DUST AROUND 4.6 BILLION YEARS AGO. THE PLANETS AND COUNTLESS MINOR BODIES THAT ORBIT THE SUN WERE FORMED FROM ACCUMULATIONS OF GAS AND DUST.

The distances from the Sun at which planets formed largely control their compositions, with the ices of water, methane, and other molecules becoming more dominant in the planets far from our star's warmth. The Solar System is much larger than just the orbits of the planet, continuing far beyond Neptune. Beyond the eighth planet lies a scattered disk of icy worlds, some of which are hundreds of miles across. Further than that is thought to lie a spherical cloud of small icy objects, known as the Oort Cloud.

▽ The orbits of the planets

All the planets follow stable paths around the Sun. These paths are almost circular and are close to lying in a flat disk. The speed of the planets' motion depends on their distances from the Sun, so Mercury travels far more rapidly than Neptune. Comets, trans-Neptunian objects, and many asteroids follow more elliptical orbits, during which their distance from the Sun varies considerably. Some long-period comets follow extremely elongated paths that take them from very close to the Sun to vast distances away over periods of thousands of years.

